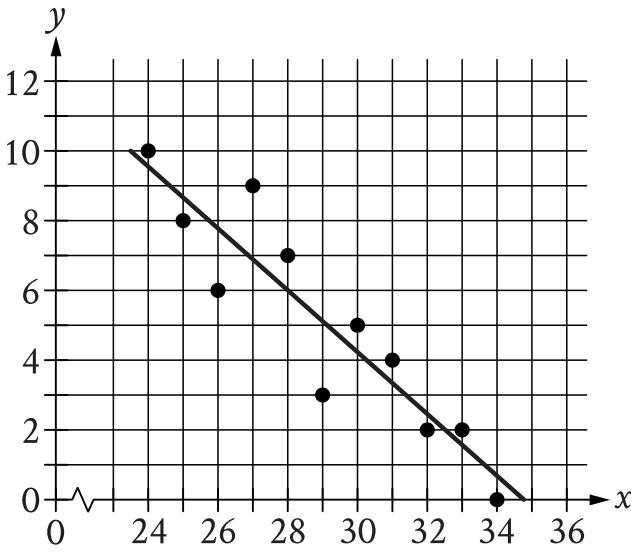
Question ID: fdfc90e4

Question

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown.



At $x = 32$, which of the following is closest to the *y*-value predicted by the line of best fit?

Answer

A. 0.4

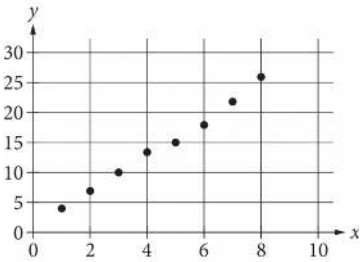
B. 1.5

C. 2.4

D. 3.3

Question ID: 9eb896c5

Question



Which of the following could be the equation for a line of best fit for the data shown in the scatterplot above?

Answer

A. $y = 3x + 0.8$

B. $y = 0.8x + 3$

C. $y = -0.8x + 3$

D. $y = -3x + 0.8$

Question ID: ab7740a8

Question

In which of the following tables is the relationship between the values of x and their corresponding y-values nonlinear?

Answer

A.

x	1	2	3	4
y	8	11	14	17

B.

x	1	2	3	4
y	4	8	12	16

C.

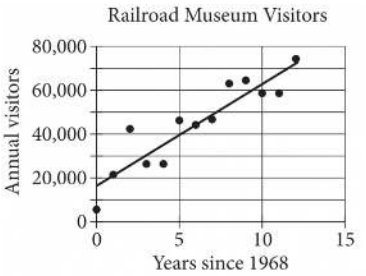
x	1	2	3	4
y	8	13	18	23

D.

x	1	2	3	4
y	6	12	24	48

Question ID: 3c5b19ef

Question



The scatterplot above shows the number of visitors to a railroad museum in Pennsylvania each year from 1968 to 1980, where t is the number of years since 1968 and n is the number of visitors. A line of best fit is also shown. Which of the following could be an equation of the line of best fit shown?

Answer

A. $n = 16{,}090 + 4{,}680t$

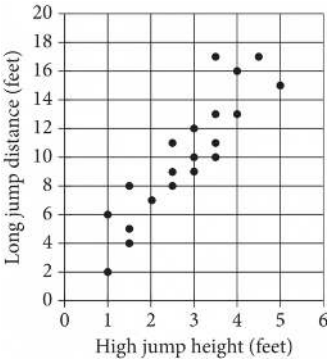
B. $n = 4{,}690 + 16{,}090t$

C. $n = 16{,}090 + 9{,}060t$

D. $n = 9{,}060 + 16{,}090t$

Question ID: 3d985614

Question



Each dot in the scatterplot above represents the height x, in feet, in the high jump, and the distance y, in feet, in the long jump, made by each student in a group of twenty students. The graph of which of the following equations is a line that most closely fits the data?

Answer

A. $y = 0.82x + 3.30$

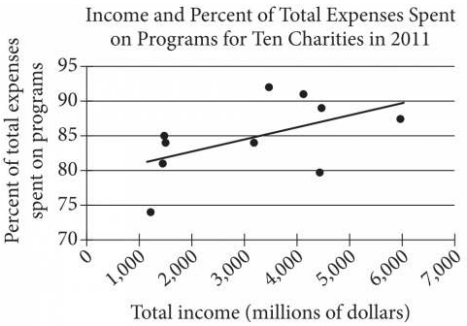
B. $y = 0.82x - 0.82$

C. $y = 3.30x + 0.82$

D. $y = 3.30x - 3.30$

Question ID: 7fd284ac

Question



The scatterplot above shows data for ten charities along with the line of best fit. For the charity with the greatest percent of total expenses spent on programs, which of the following is closest to the difference of the actual percent and the percent predicted by the line of best fit?

Answer

A. 10%

B. 7%

C. 4%

D. 1%

Question ID: 7ac5d686

Question

An inspector begins a day of work with a large sample of shirts that need to be checked for defects. The inspector works at a constant rate throughout the morning. What type of model is best to model the number of shirts remaining to be checked for defects at any given time throughout the morning?

Answer

A. A linear model with a positive slope

B. A linear model with a negative slope

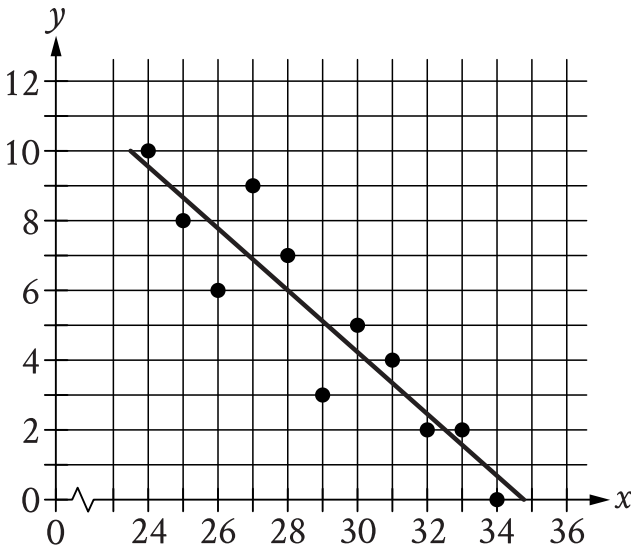
C. An exponential growth model

D. An exponential decay model

Question ID: f46139df

Question

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown.



At $x = 25.5$, which of the following is closest to the *y*-value predicted by the line of best fit?

Answer

A. 6.2

B. 7.3

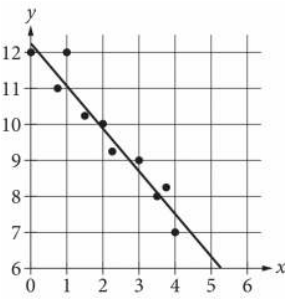
C. 8.2

D. 9.1

Question ID: 1adb39f0

Question

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown. Which of the following is closest to the difference between the y-coordinate of the data point with $x = 1$ and the y-value predicted by the line of best fit at $x = 1$?



Answer

A. 1

B. 2

C. 5

D. 12

Question ID: 9a144a01

Question

Which of the following is true about the values of 2^x and $2x+2$ for $x>0$?

Answer

A. For all $x>0$, it is true that $2^x<2x+2$.

B. For all $x>0$, it is true that $2^x>2x+2$.

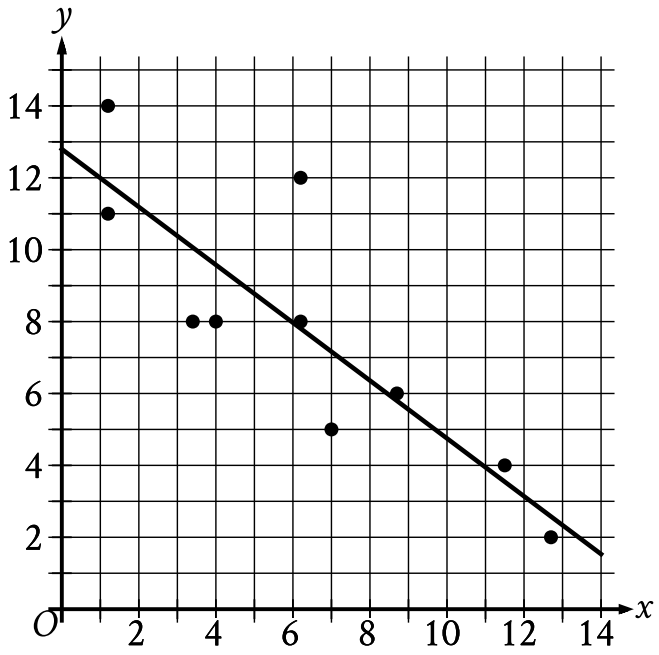
C. There is a constant c such that if $0<x<c$, then $2^x<2x+2$, but if $x>c$, then $2^x>2x+2$.

D. There is a constant c such that if $0<x<c$, then $2^x>2x+2$, but if $x>c$, then $2^x<2x+2$.

Question ID: 03a16790

Question

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



Which of the following is closest to the slope of the line of best fit shown?

Answer

A. -2.4

B. -0.8

C. 0.8

D. 2.4